# DATA ENGINEERING — STUDY NOTES — 1

## 1. Database vs Data Warehouse vs Data Lake vs Delta Lake

### Why do we need different storage systems?

Imagine an online shopping company. Customers register, place orders, make payments, upload product images and browse products. The company must store all this information, but it uses the data for different jobs: daily transactions, business reporting and raw-data exploration. Different storage technologies are designed for these different needs.

### Database — what is it?

A database is an organized system for storing, retrieving, updating and deleting information. Applications use databases for everyday operations. Think of it as a digital register that an application can work with.

### Example 1: Banking application

When you transfer ₹1,000 to a friend, the banking application checks your balance, deducts the money from your account, adds it to your friend's account and records the transaction. The database stores and manages these records.

### Example 2: College management system

A college stores student IDs, names, departments, marks, attendance and fee details. When a student changes a mobile number, the application updates the existing student record.

Example SQL:

```
SELECT * FROM Students;

UPDATE Students
SET Marks = 95
WHERE Student_ID = 102;
```

OLTP (Online Transaction Processing) is a common database workload. It handles frequent day-to-day operations such as ATM withdrawals, order placement, ticket booking and payment recording. Examples: MySQL, PostgreSQL, SQL Server, Oracle Database and MongoDB.

### Data Warehouse — what is it?

A Data Warehouse is a centralized system that collects data from different sources, prepares it and stores it for reporting, analytics and business decision-making. It is useful for studying trends and historical information.

### Example 1: E-commerce sales analysis

A retailer may want to know which product generated the most revenue, which city had the highest sales, or which month performed best. Data from orders, payments and customer systems can be combined for analysis.

### Example 2: Hospital analytics

A hospital can combine information from registration, appointments, billing and pharmacy systems to analyze yearly patient counts, common treatments, average costs and monthly admissions.

## ETL and ELT

| Term | Meaning |
|---|---|
| ETL | Extract data from sources, Transform or clean it, then Load it into the target. |
| ELT | Extract data, Load it into the target platform, then Transform it there. |

```
Source systems
      |
   ETL / ELT
      |
 DATA WAREHOUSE
      |
 Reports and BI dashboards
```

OLAP (Online Analytical Processing) is a common analytical workload. It supports complex queries across large amounts of historical or integrated data. Examples: Snowflake, Google BigQuery, Amazon Redshift and Azure Synapse Analytics.

## Data Lake — what is it?

A Data Lake is a storage environment that can hold large volumes of data in its original or near-original form. It can store structured, semi-structured and unstructured data. Data can be cleaned and shaped later for analytics, machine learning or other processing.

| Data type | Meaning | Example |
|---|---|---|
| Structured | Organized in a defined table-like structure | CSV with customer_id, name and city |
| Semi-structured | Flexible organization with labels or keys | JSON events, XML, application logs |
| Unstructured | Does not naturally fit into rows and columns | Images, audio recordings, videos |

## Example 1: Video platform

A video platform creates user records, video metadata, comments, viewing events, application logs and video files. A Data Lake can store these different data formats for later processing and analysis.

## Example 2: Healthcare

A hospital may store patient data extracts, medical reports, X-ray images, MRI scans and recordings. A Data Lake can keep these varied files together in scalable storage, subject to privacy and access controls.

Schema-on-read is commonly associated with Data Lakes: structure can be applied when data is read or processed, rather than requiring every raw file to fit a predefined relational table before storage. Examples of storage services used for data lakes include Azure Data Lake Storage, Amazon S3 and Google Cloud Storage.

## Delta Lake — what is it?

Delta Lake is an open-source storage framework that adds reliability and table-management features to data stored in a Data Lake. A Delta table uses data files together with a transaction log that records table changes.

### Example 1: Bank transaction data

A data engineering team receives new bank transaction records every day. Delta Lake helps manage writes reliably and provides transaction history for the table.

### Example 2: E-commerce customer updates

A customer originally lives in Chennai and later moves to Nellore. A pipeline can use MERGE to update the existing customer row when a matching ID is found, or insert a row when the ID is new.

Important Delta Lake features:

- ACID transactions — support reliable table changes.
- Schema enforcement — checks incoming data against the table schema.
- Schema evolution — can allow supported schema changes, such as adding a column, when configured.
- Time Travel — allows queries against earlier table versions while the required history and data files remain available.
- MERGE — supports update-and-insert patterns, often called upsert.

```
MERGE INTO customers AS target
USING new_customers AS source
ON target.id = source.id
WHEN MATCHED THEN
  UPDATE SET target.city = source.city
WHEN NOT MATCHED THEN
  INSERT (id, name, city)
  VALUES (source.id, source.name, source.city);
```

Delta Lake is not a cloud storage account by itself. It works with storage such as Azure Data Lake Storage or Amazon S3, and is commonly used with Apache Spark and Azure Databricks.

### Quick comparison

| Feature | Database | Data Warehouse | Data Lake | Delta Lake |
|---|---|---|---|---|
| Purpose | Run applications | Analyze business data | Store raw datasets | Reliable tables on lake storage |
| Typical data | Operational records | Prepared, integrated data | Structured, semi-structured, unstructured | Table data plus transaction log |
| Use | Orders and payments | Reports and trends | Data science and exploration | Pipelines, updates, version history |
| Workload | Often OLTP | Often OLAP | Storage and processing | Reliable analytical processing |

Memory aid: Database = operate; Warehouse = analyze; Lake = store raw data; Delta Lake = manage lake data as reliable tables.

## 2. What is a Transaction in Database Terms?

A transaction is one logical unit of work made up of one or more database operations. The operations are treated as a group: the system commits the successful unit or rolls back its uncommitted changes if the work cannot be completed.

### Example 1: Bank transfer

Account A has ₹5,000 and Account B has ₹2,000. You transfer ₹1,000. The database must subtract ₹1,000 from A and add ₹1,000 to B. Both changes belong to the same transaction so the transfer is not left half-completed.

```
BEGIN TRANSACTION;

UPDATE Accounts
SET balance = balance - 1000
WHERE account_id = 1;

UPDATE Accounts
SET balance = balance + 1000
WHERE account_id = 2;

COMMIT;
```

If the work fails before it is committed, the transaction can be rolled back:

```
ROLLBACK;
```

### Example 2: Online shopping

When a customer buys a phone, the system may need to create an order, record payment and reduce inventory. These related operations must be coordinated so a failure does not leave inconsistent business records.

| Command | Purpose |
|---|---|
| BEGIN TRANSACTION | Starts a transaction (syntax varies by database). |
| COMMIT | Makes transaction changes permanent. |
| ROLLBACK | Undoes uncommitted changes in the transaction. |

The SQL example uses SQL Server-style transaction syntax. Other database systems may use slightly different syntax or behavior.

## 3. What are ACID Properties (with Examples)?

ACID describes four properties that help database transactions remain reliable: Atomicity, Consistency, Isolation and Durability.

| Property | Simple meaning | Bank-transfer example |
|---|---|---|
| Atomicity | All or nothing | Both debit and credit happen, or neither change is kept. |
| Consistency | Rules remain valid | The transaction respects constraints and business rules, such as valid account balances. |
| Isolation | Concurrent transactions do not interfere incorrectly | Two transfers at once should not create an invalid result. |
| Durability | Committed changes survive failures | After commit, the transfer remains recorded even if the server restarts. |

### Atomicity

A transaction is treated as one indivisible unit. It should not leave only half of the intended work applied.

### Consistency

A committed transaction moves the database from one valid state to another, respecting defined constraints and business rules.

### Isolation

Isolation controls how concurrent transactions interact. It aims to prevent unsafe interference. Database systems provide different isolation levels, balancing correctness and performance.

### Durability

Once the database confirms COMMIT, the committed result is designed to survive a crash or restart, using mechanisms such as transaction logs and persistent storage.

Memory aid: A = All or nothing; C = Correct rules; I = Independent concurrent work; D = Data stays after commit.

## 4. What is Time Travel in Delta Table?

Time Travel is a Delta Lake feature that lets you read an earlier version of a Delta table. The Delta transaction log records table operations and versions. This can help investigate mistakes, compare versions or reproduce an earlier result.

### Example 1: Customer table changes

| Version | Change | Illustrative row count |
|---|---|---|
| 0 | Initial table created | 100 |
| 1 | Added 20 customers | 120 |
| 2 | Updated some customer cities | 120 |
| 3 | Deleted 5 test records | 115 |

If the current table is version 3, you can query version 1 to inspect the earlier state, provided the required log history and data files are still available.

### Example 2: Pipeline mistake

Suppose a Databricks pipeline accidentally overwrites a table with incorrect values. You can inspect an earlier version with Time Travel to compare the data and investigate what changed. Time Travel is useful for investigation; recovery actions should be planned carefully.

View table history:

```
DESCRIBE HISTORY customers;
```

Read a particular version:

```
SELECT *
FROM customers VERSION AS OF 1;
```

Read the table as of a timestamp (replace with a real timestamp from table history):

```
SELECT *
FROM customers TIMESTAMP AS OF '2026-11-04T10:00:00';
```

Important: Time Travel is not a permanent backup. It depends on retention settings and the availability of the underlying data files and transaction log. VACUUM can remove old data files beyond the retention

period, making older versions unavailable. Follow your team's retention and safety policies before running VACUUM.

## 5. How these concepts connect in Azure Databricks

One common Data Engineering workflow looks like this:

```
APPLICATION DATABASES / SOURCE FILES
                |
                v
        AZURE DATA LAKE STORAGE
          (raw files and data)
                |
                v
        AZURE DATABRICKS + SPARK
       (read, clean, join, transform)
                |
                v
             DELTA TABLES
       (reliable processed datasets)
                |
                v
       REPORTING / ANALYTICS / BI
```

For example, a team may land a CSV file in Azure Data Lake Storage, read it with Databricks, clean it using PySpark, write the result as a Delta table, query it with SQL and use the prepared data for reporting. This is a common pattern, not a mandatory architecture.

## 6. Final revision in one minute

A Database supports day-to-day application work. A Data Warehouse supports business analysis. A Data Lake stores large amounts of raw data in many formats. Delta Lake adds reliable table operations and version history to data-lake storage. A transaction is a logical unit of database work. ACID describes transaction reliability. Delta Lake Time Travel lets you inspect earlier table versions while the required history and data files are retained.